Multi-Omics Predictive Modelling for Breast Cancer Relapse

Benchmarking against clinical staging, with external validation · 5-year relapse endpoint
Generated 2026-08-31 20:36 · seed 42

Abstract

Headline result. On identical outer folds, the clinical baseline performed best. Model A (clinical: age, stage, PAM50) reached AUC **0.689** [0.669-0.708]; Model B (XGBoost on clinical + expression + methylation) reached **0.625** [0.608-0.643]; Model C (pathway-informed neural network) reached **0.619** [0.602-0.637]. Adding ~19,000 omics features did not improve on a two-variable clinical model at this sample size.

Cohort. n = 254 patients with matched expression, methylation and a usable label; 94 relapse-positive (37.0%). 744 patients were excluded for censoring before the 5-year horizon.

External validation. METABRIC n = 1835 (495 positive). Best expression-only AUC **0.6165** (panel 50, logreg_l2). METABRIC has no matched 450k methylation, so the methylation half of the signature could not be tested externally.

What this report does and does not claim

This is a negative-leaning result reported as found. The pipeline was pre-specified: feature selection is nested inside every cross-validation split, models are compared on identical fold indices, and no model was retuned after seeing its comparison. Where the data could not support the planned analysis, the deviation is stated rather than worked around.

1. Data and cohort construction

1.1 Sources (Gate 0)

Dataset	Rows	Cols	Size
tcga_expression	60,660	1,227	138.5 MB
tcga_methylation	486,427	894	2.99 GB
tcga_clinical	1,255	85	0.3 MB
tcga_survival	1,232	4	0.0 MB
tcga_cdr	12,591	34	2.4 MB
tcga_subtype	7,734	9	0.0 MB
methyl_manifest	486,428	33	64.0 MB
mcp_genes	111	4	0.0 MB
gencode_gtf	3,014,859	1	44.5 MB
metabric_expr	20,603	1,982	689.3 MB
metabric_patient	2,513	24	0.4 MB
metabric_sample	2,513	13	0.3 MB

Full URLs and md5 checksums are in *results/tables/gate0_data_manifest.csv*.

1.2 Label definition

Relapse is defined from the Progression-Free Interval (PFI) in the TCGA-CDR curated endpoints (Liu et al., Cell 2018), at a 1825-day (5-year) horizon:

PFI event	PFI time	Label
1	≤ 1825d	positive (relapsed)
0	≥ 1825d	negative (relapse-free)
0	< 1825d	**EXCLUDED** - censored early, status unknown
1	> 1825d	negative

The third row is the one that is easy to get wrong. A patient censored at two years is not a negative; their five-year status is unknown. Excluding them is correct and expensive: it is the single largest filter in this pipeline.

1.3 Horizon choice (deviation from spec)

1.4 Filtering accounting (Gate 1)

Step	Unit	In	Out	Dropped
TCGA-CDR rows (all cancer types)	samples	12,591	12,591	
restrict to BRCA	samples	12,591	1,236	11,355
drop TCGA-redacted patients	samples	1,236	1,230	6
de-duplicate to one row per patient	samples	1,230	1,092	138
drop patients with missing PFI or PFI.time	samples	1,092	1,091	1
EXCLUDE censored before 1825d (status unknown)	samples	1,091	347	744
positives (relapse <= 5y)	patients	123	123	
negatives (relapse-free through 5y)	patients	224	224	
expression: raw genes	genes	60,660	60,660	
expression: raw samples (aliquots)	aliquots	1,226	1,226	
expression: keep primary tumour (sample type 01)	aliquots	1,226	1,106	120
expression: collapse aliquots to one per patient	aliquots	1,106	1,095	11
expression: drop chrY PAR_Y duplicate accessions	genes	60,660	60,616	44
expression: genes mappable to GENCODE v36	genes	60,616	60,616	
expression: keep protein_coding	genes	60,616	19,944	40,672
expression: collapse duplicate gene symbols (mean)	genes	19,944	19,938	6
expression: drop zero-variance genes	genes	19,938	19,516	422
expression: drop bottom 50% by variance	genes	19,516	9,758	9,758
450k manifest probes	probes	486,428	486,428	
manifest: drop chrX/chrY probes	probes	486,428	474,780	11,648
manifest: probes with a RefGene annotation	probes	356,119	356,119	
manifest: promoter-associated probes (1stExon,5'UTR,TSS1500,TSS200)	probes	356,119	192,998	163,121
methylation: probes in file	probes	486,427	192,998	293,429
methylation: keep primary tumour (sample type 01)	aliquots	893	791	102
methylation: collapse aliquots to one per patient	aliquots	791	784	7
methylation: drop probes with >10% NA	probes	192,998	166,054	26,944
methylation: drop samples with >10% NA	patients	784	784	
methylation: aggregate promoter probes to gene mean beta	genes	181,896	19,146	162,750
methylation: drop bottom 50% by variance	genes	19,146	9,573	9,573
patients with a usable label	patients	347	347	
patients with expression	patients	1,095	1,095	
patients with methylation	patients	784	784	
INTERSECT label AND expression AND methylation	patients	347	254	93
genes shared by both modalities	genes	5,669	5,669	

Final cohort: n = 254, 94 positive / 160 negative (37.0% prevalence), 9758 expression features, 9573 methylation features, 5669 genes measured by both modalities.

2. Model comparison

All three models are evaluated on the SAME outer fold indices (5-fold x 10 repeats = 50 folds), loaded from a written fold file rather than regenerated from a seed, so the comparison is genuinely paired.

Model	Description	AUC (95% CI)	AP (95% CI)
A	Logistic regression, clinical only (age, stage, PAM50)	0.689 [0.669, 0.708]	0.632 [0.611, 0.652]
B	XGBoost, clinical + expression + methylation	0.625 [0.608, 0.643]	0.545 [0.524, 0.566]
C	Pathway-informed neural network (P-NET style)	0.619 [0.602, 0.637]	0.538 [0.516, 0.559]

2.1 Paired tests on identical folds

Comparison	Mean AUC difference	Paired t p	Wilcoxon p	Folds where 2nd wins
A vs B	-0.0635	2.19e-07	1.87e-07	9/50
A vs C	-0.0697	1.11e-06	3.59e-06	13/50
B vs C	-0.0062	3.88e-01	3.14e-01	21/50

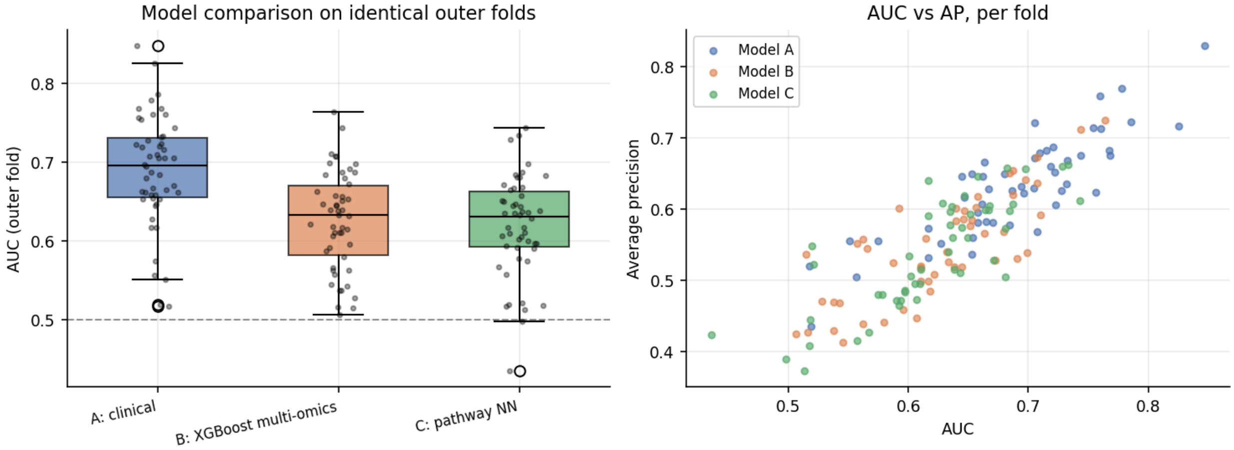

Figure 2. Per-fold AUC by model on identical outer folds.

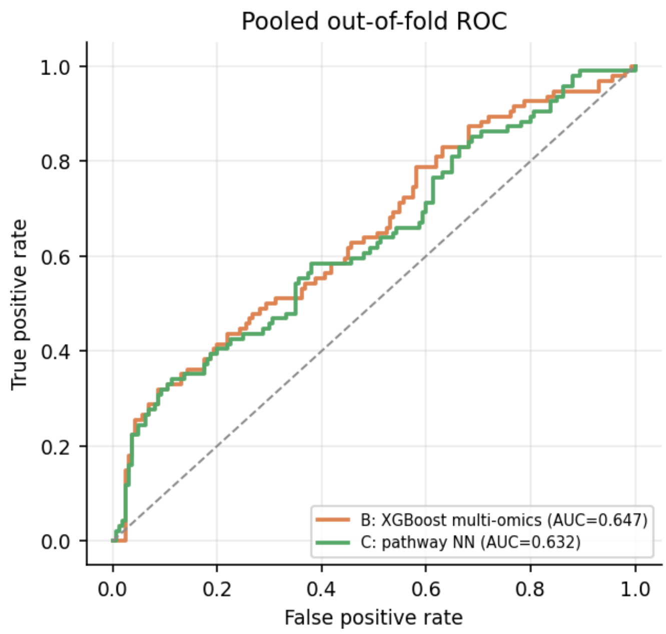

Figure 3. Pooled out-of-fold ROC curves.

2.2 Gate 2 diagnostic: why is the clinical baseline strong?

Model A exceeded the 0.60-0.70 band the spec expects for clinical-only BRCA models, so it was investigated before proceeding. It is not leakage: PAM50 (which is expression-derived, and so arguably should not sit in a 'clinical' baseline at all) contributes almost nothing, and the signal is carried by pathologic stage.

Variant	AUC (95% CI)	AP
A_full (age+stage+PAM50)	0.689 [0.669, 0.708]	0.632
A_no_PAM50 (age+stage only, TRUE clinical)	0.702 [0.680, 0.724]	0.654
A_PAM50_only	0.550 [0.533, 0.567]	0.411
A_age_only	0.594 [0.577, 0.612]	0.506
A_stage_only	0.666 [0.647, 0.684]	0.527
A_full_excl_stageIV	0.654 [0.634, 0.674]	0.555
A_no_PAM50_excl_stageIV	0.668 [0.646, 0.691]	0.581

Stage composition of the cohort:

Stage	n	Relapse rate
missing	10	60.0%
1	39	15.4%
2	126	30.2%
3	70	50.0%
4	9	100.0%

The stage gradient is steep, and the landmark labelling rule sharpens it further: requiring survivors to have full follow-up preferentially retains lower-stage patients as negatives, while early events remain as positives. The strong clinical baseline is a real property of this selected cohort, not an artefact of the code.

3. Feature ranking (Gate 3)

#	Feature	Block	Mean |SHAP|	Fold selection
1	TSPYL5	expression	0.2842	2%
2	CCL7	methylation	0.2017	90%
3	ZNF135	methylation	0.1844	2%
4	EPHA5	expression	0.1700	2%
5	SLC22A16	methylation	0.1657	2%
6	SCG2	expression	0.1071	42%
7	A2M	methylation	0.1059	2%
8	EMCN	methylation	0.1031	6%
9	MIR1295	methylation	0.1021	2%
10	VENTX	expression	0.1007	4%
11	RIMBP3	expression	0.0929	10%
12	GNA15	methylation	0.0910	6%
13	GRIP1	methylation	0.0899	54%
14	MSX1	expression	0.0887	2%
15	CADPS	expression	0.0874	32%
16	CDK5R1	expression	0.0873	78%
17	C5orf63	expression	0.0859	2%
18	SAMD11	expression	0.0857	16%
19	LRRC43	expression	0.0821	2%
20	ZNF257	methylation	0.0818	6%
21	CCDC83	expression	0.0806	2%
22	GRIK2	expression	0.0805	36%
23	KCNK1	methylation	0.0792	100%
24	MIR135A1	methylation	0.0739	10%
25	LILRA1	methylation	0.0725	4%
26	LXN	methylation	0.0716	8%
27	ANKRD18B	expression	0.0698	2%
28	FLJ40504	methylation	0.0687	52%
29	H2BU1	expression	0.0684	24%
30	NTS	methylation	0.0682	2%

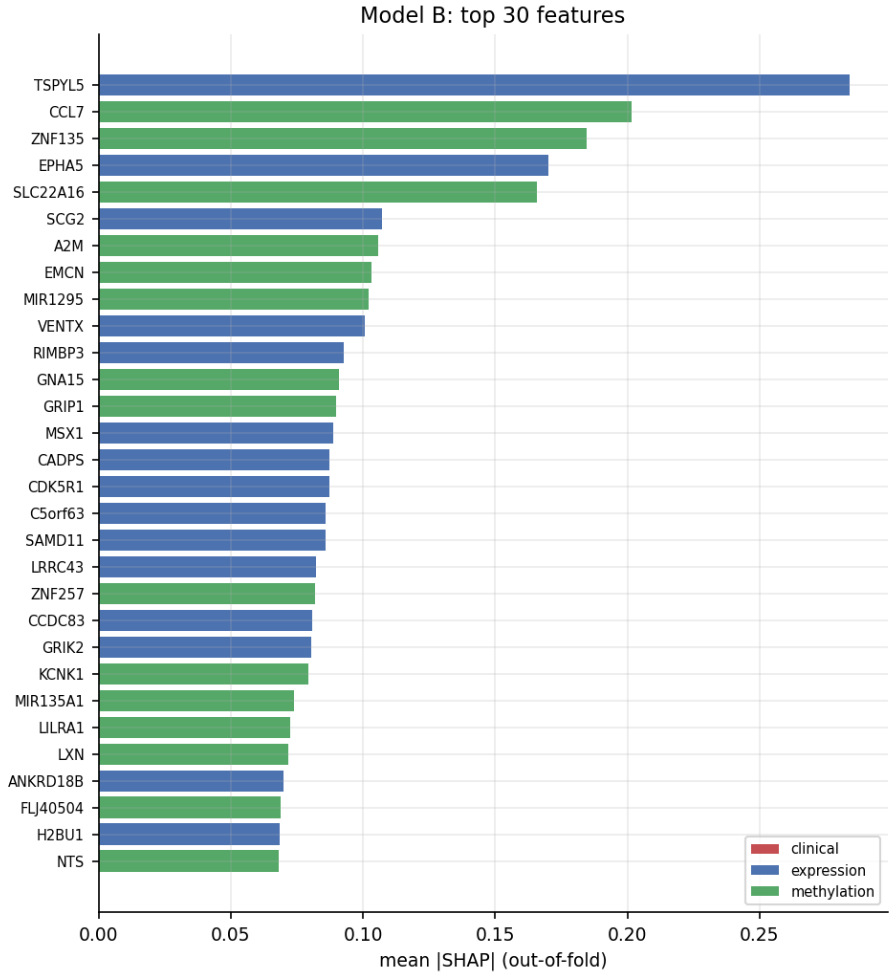

Figure 4. Model B top 30 features by out-of-fold mean |SHAP|.

4. Signature reduction (Gate 5)

Panel size	AUC (95% CI)	AP
20	0.646 [0.626, 0.665]	0.548
50	0.639 [0.621, 0.656]	0.546
100	0.634 [0.617, 0.650]	0.535
250	0.634 [0.616, 0.651]	0.543
500	0.631 [0.614, 0.648]	0.542

Performance plateaus at **20** markers (smallest panel within one standard error of the best mean AUC, which was 0.646 at 20 markers). 50-marker panel composition: {'meth': 40, 'expr': 9, 'clin': 1}.

Panels are re-selected inside every fold from the training half only; the reported marker list is the consensus across folds (selection frequency is given per marker in *results/tables/signature_panels.csv*). Scoring a globally-chosen panel would have leaked the test folds into the panel definition.

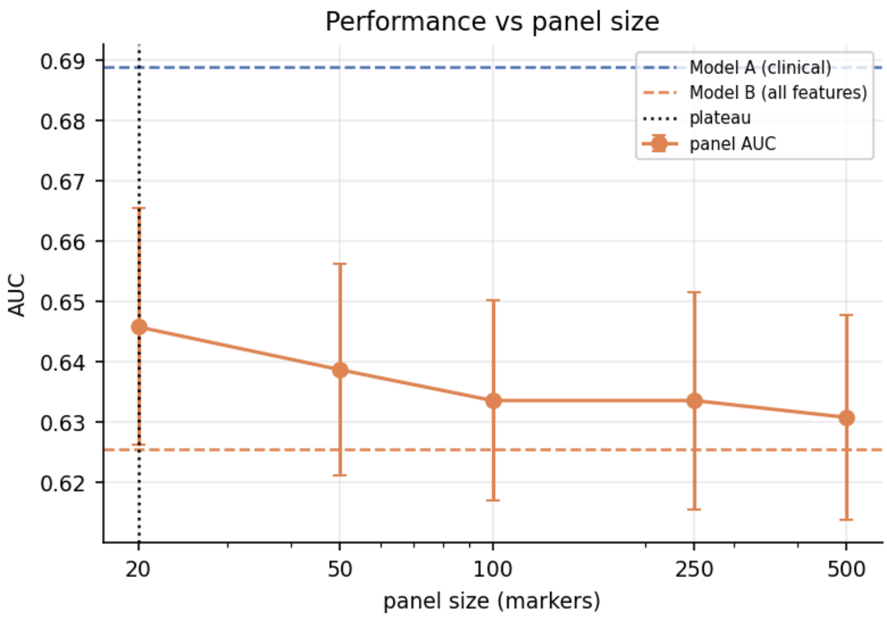

Figure 5. AUC against panel size, with model baselines.

4.1 The 50-marker panel

#	Marker	Block	Freq.	#	Marker	Block	Freq.
1	stage	clin	100%	26	LRRC2	expr	44%
2	MIR1284	meth	100%	27	CAPN13	meth	44%
3	MIR570	meth	100%	28	TNNT2	meth	42%
4	BGLAP	meth	96%	29	IL17REL	meth	42%
5	LIMCH1	meth	92%	30	C14orf49	meth	40%
6	HIST1H2BK	meth	86%	31	FAM71F2	meth	36%
7	MIR600	meth	80%	32	C9orf131	meth	36%
8	MICALCL	meth	80%	33	MAPK8IP2	meth	36%
9	LOC285045	meth	80%	34	BTG2	expr	34%
10	CCDC89	meth	78%	35	SPINK7	meth	34%
11	DGUOK	meth	64%	36	GHRL	meth	32%
12	ATF7	meth	64%	37	ACSF2	expr	32%
13	FLT3	expr	64%	38	QPRT	meth	32%

14	ASAP1IT1	meth	62%	39	AVIL	meth	30%
15	CALML5	meth	58%	40	ABCA1	meth	30%
16	AOC2	meth	58%	41	SNORA71B	meth	30%
17	SSC5D	meth	58%	42	CLRN1OS	meth	30%
18	ZNF710	meth	58%	43	RCOR2	expr	30%
19	EIF3G	meth	54%	44	LEF1	expr	30%
20	CLEC9A	expr	54%	45	NDUFB6	meth	30%
21	SLC38A8	meth	52%	46	LOC730755	meth	30%
22	ASIC1	expr	50%	47	SPRED2	meth	30%
23	EXPH5	meth	48%	48	F10	meth	28%
24	CCDC9B	expr	48%	49	A2ML1	meth	28%
25	ATP6V0A4	meth	44%	50	RNASE6	meth	28%

5. Immune deconvolution (Gate 6)

Method deviation, stated plainly. The spec asks for immunedeconv (R) with quanTIseq or MCP-counter. The immunedeconv install failed on this machine: Bioconductor dependencies (limSolve, biomaRt, sva, xCell, ConsensusTME, ComICS, quantiseqr) were unavailable. MCP-counter was therefore reimplemented directly in Python from the published 111-transcript, 10-population marker set (Becht et al., Genome Biology 2016) - the same signature the R package wraps. MCP-counter is a marker-gene aggregate, not a fitted deconvolution, so the reimplementation is faithful rather than approximate. quanTIseq was NOT run: it requires the TIL10 signature matrix that ships with the R package.

Patients were split at the median out-of-fold risk score from Model B (AUC 0.625): 127 high-risk vs 127 low-risk. Observed relapse rate 44.1% vs 29.9%.

Population	High risk	Low risk	Delta	p	FDR
Endothelial cells	8.773	9.284	-0.511	8.91e-05	**0.000891**
Myeloid dendritic cells	5.623	6.487	-0.864	0.000322	**0.00161**
B lineage	5.595	6.471	-0.876	0.000615	**0.00205**
NK cells	2.688	3.210	-0.521	0.00108	**0.0027**
Cytotoxic lymphocytes	4.901	5.401	-0.500	0.00309	**0.00551**
Neutrophils	5.522	5.843	-0.320	0.0033	**0.00551**
T cells	6.244	6.832	-0.588	0.00418	**0.00597**
CD8 T cells	6.528	7.084	-0.556	0.0204	**0.0254**
Fibroblasts	13.958	14.210	-0.252	0.169	0.188
Monocytic lineage	9.708	9.796	-0.088	0.563	0.563

Gate 6 verdict. The signature separates risk groups that ALSO differ in immune infiltration -- it is at least partly an immune readout.

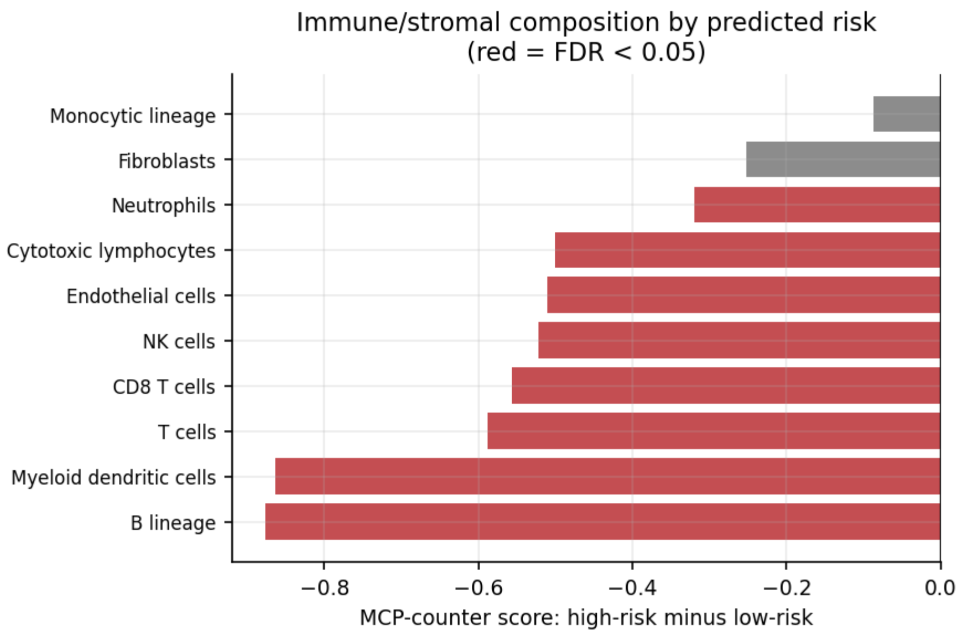

Figure 6. MCP-counter scores by predicted risk group.

6. External validation in METABRIC (Gate 7)

Known constraint, not worked around. METABRIC has no matched Illumina 450k methylation, so only the expression component of the signature can be tested externally. cBioPortal does host a promoter-RRBS methylation profile for METABRIC, but RRBS on a sample subset is a different assay and is not a substitute; it was not used. Transfer is also cross-platform (TCGA RNA-seq to Illumina HT-12 microarray), so genes were z-scored within each cohort independently, and degradation is expected.

Panel	Expr. markers	Mapped to METABRIC	Rate	Methylation markers (untestable)
20	4	4	100%	15
50	9	9	100%	40
100	27	26	96%	72
250	80	77	96%	169
500	186	179	96%	313

Panel	Model	Genes used	METABRIC AUC	METABRIC AP
50	logreg_l2	9	**0.617**	0.354
50	xgboost	9	**0.582**	0.338
100	logreg_l2	26	**0.578**	0.331
100	xgboost	26	**0.570**	0.340
250	logreg_l2	77	**0.573**	0.318
250	xgboost	77	**0.611**	0.362
500	logreg_l2	179	**0.559**	0.300
500	xgboost	179	**0.605**	0.356

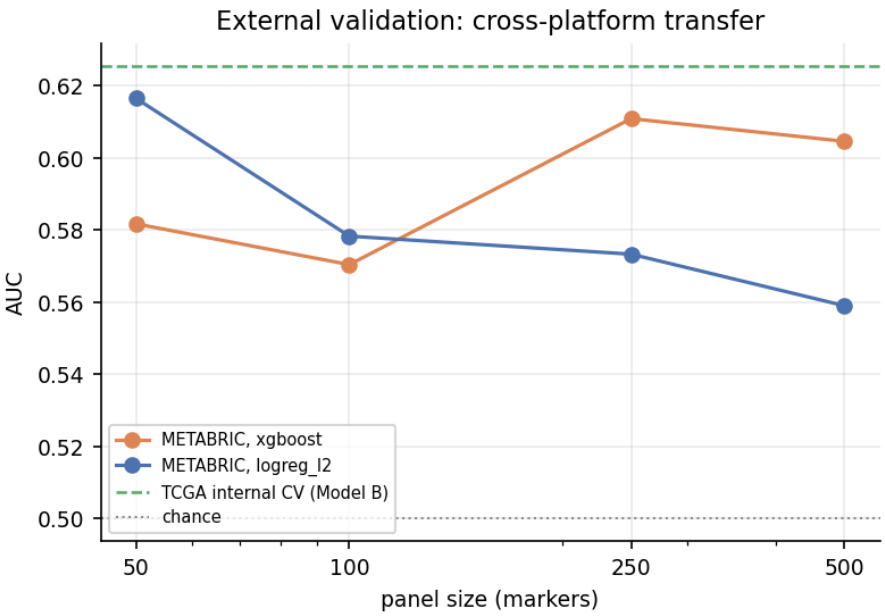

Figure 7. External validation across panel sizes.

Kaplan-Meier by predicted risk group. TCGA (out-of-fold risk): log-rank p = 0.0213; METABRIC (external): log-rank p = 1.27e-05.

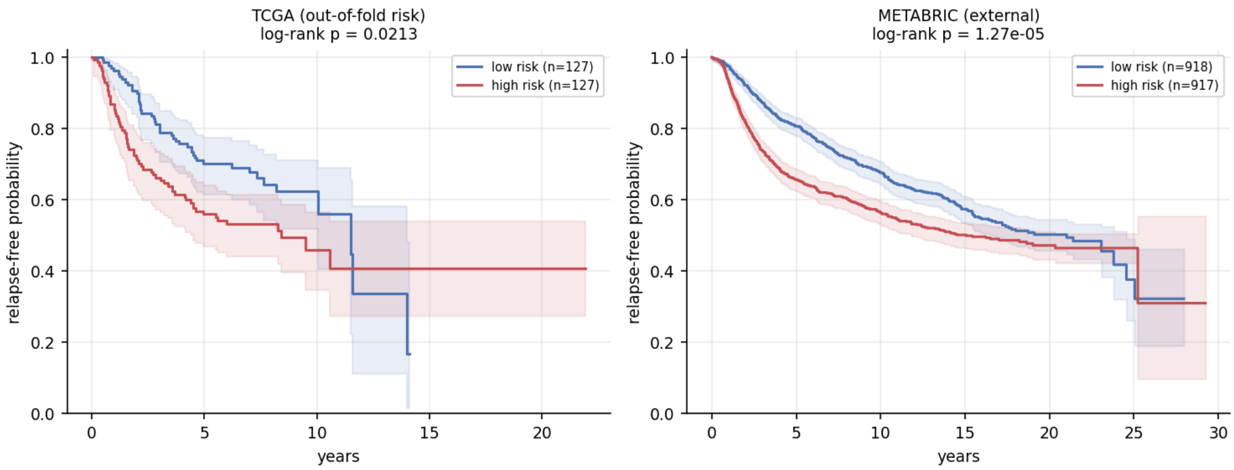

Figure 8. Relapse-free survival by predicted risk group. Note that negatives are required by the landmark rule to have full follow-up, which flattens the early part of the curves.

7. Limitations

• **Sample size.** n = 254 with ~19,000 candidate features. This is the dominant limitation and the most likely reason the omics models do not beat the clinical baseline. The spec's n = 550-750 target is not reachable in TCGA-BRCA once matched methylation and the early-censoring rule are both enforced.

• **Early-censoring exclusion.** 744 patients were discarded because their status at the horizon is genuinely unknown. This is the methodologically correct choice, but it induces a selection effect: negatives are, by construction, patients with long follow-up. Prevalence and the strength of stage as a predictor are both inflated relative to an unselected cohort.

• **No matched external methylation.** Only the expression half of the signature was externally validated. A multi-omics signature validated on one modality is a partially validated signature.

• **Cross-platform transfer.** RNA-seq to microarray transfer degrades performance independently of biology; the METABRIC number confounds signature quality with platform shift.

• **PAM50 in the 'clinical' baseline.** PAM50 is expression-derived. It is included because the spec lists it, but it makes Model A not a purely clinical model. The diagnostic in section 2.2 quantifies its (small) contribution.

• **Grade unavailable.** Histologic grade is 100% missing in the TCGA-CDR fields for this cohort and was dropped, so the clinical baseline is thinner than specified.

• **quanTIseq not run**, and immunedeconv (R) not used - see section 5.

• **Single cohort discovery.** Discovery is entirely within TCGA-BRCA; no multi-cohort discovery or meta-analysis was attempted.

8. Reproducibility

Item	Value
Random seed	42
Outer CV	5-fold x 10 repeats (50 folds), shared across models via cv_folds.json
Inner CV	3-fold, 10 random search candidates
Leakage control	All selection/imputation/scaling inside sklearn Pipelines, refit per split
Container	docker/Dockerfile (python:3.11-slim, pinned)
Workflow	nextflow/main.nf (DSL2), profiles: standard, docker, conda, smoke
Smoke test	scripts/smoke_test.sh - full pipeline on a 100-patient fixture
Environment	env/requirements.txt (pip freeze), env/requirements-docker.txt

Every script takes *--config*, writes to *results/*, logs each filter with in/out counts, and records the seed. Figures are generated by a separate script that only reads tables, so a figure cannot disagree with the table behind it.